

Beam-Steered Single-LED Single-Photodiode Optical Wireless Positioning for 3D Indoor Localization

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Abstract—(draft) This paper investigates a practical three-dimensional optical wireless positioning (OWP) architecture in which a single ceiling-mounted LED transmitter sequentially steers its optical beam towards K distinct orientations, while a single upward-looking photodiode (PD) placed at an unknown position gathers N power samples per orientation. Two successive estimation stages are proposed: (i) direction finding of the LED–PD line-of-sight, followed by (ii) range recovery through a dedicated beam-steering orientation that maximises the received signal-to-noise ratio (SNR). We derive the Cramér–Rao lower bound, here expressed as the position-error bound (PEB), to benchmark any unbiased estimator. A genetic-algorithm optimiser is then designed to compute the K -orientation sets that minimise the 90th percentile of the PEB over a cuboidal testbed. Three estimators are analysed: a statistically efficient but iterative nonlinear (NL) estimator, a closed-form generalised least-squares (GLS) estimator, and a computationally lightweight weighted least-squares (WLS) approximation. Extensive simulations corroborate the theoretical bound and reveal that the GLS reaches near-optimal performance with two orders of magnitude lower latency than NL, whereas WLS attains within 1 dB of GLS for typical indoor geometries. Guidelines are finally provided to select K and the Lambertian half-power angle $\Phi_{1/2}$, trading off latency against accuracy.

Index Terms—3D localization, Cramér–Rao lower bound (CRLB), generalized least squares (GLS), indoor localization, light emitting diode (LED), optical wireless positioning (OWP), received signal strength (RSS), single LED

I. INTRODUCTION

GLOBAL Navigation Satellite Systems (GNSS) provide ubiquitous outdoor positioning but are unreliable indoors due to severe attenuation, blockage, and multipath, which motivates dedicated indoor positioning systems (IPS) [1]. Among IPS, radio-frequency (RF) techniques leveraging existing communications infrastructure, such as Wi-Fi [2], Bluetooth Low Energy (BLE) beacons [3], ultra-wideband (UWB) [4], Radio Frequency Identification (RFID) [5], and emerging 5G positioning [6], are widely studied and deployed [6], [7]. However, in cluttered interiors they are strongly affected by multipath, non-line-of-sight (NLOS) conditions, and time-varying interference in shared spectrum, typically yielding meter-level accuracy unless dense infrastructure, calibration, or specialized hardware is used [8], [9].

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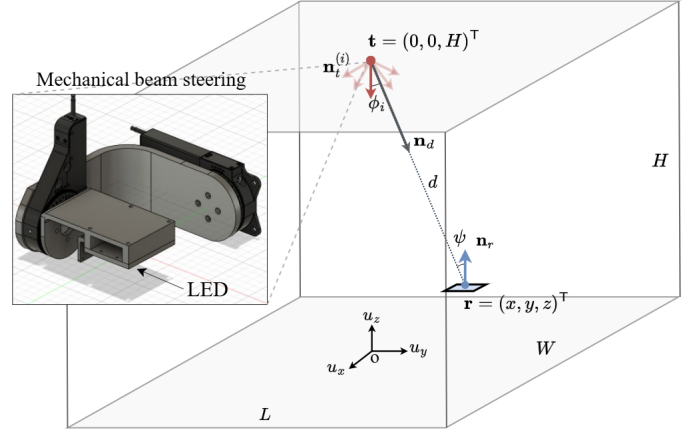


Fig. 1. Configuration of the system single-LED Optical Wireless Positioning.

In this context, Optical Wireless Positioning (OWP) leverages low-cost, small-form-factor light-emitting diodes (LEDs) that transmit modulated optical signals, often imperceptible to humans, captured by photodiodes (PDs) or cameras to transform lighting into a positioning network [10]. The spatial confinement of light (no wall penetration), immunity to RF electromagnetic interference, high spatial reuse, and more deterministic line-of-sight optical channels enable centimeter-level localization with cost-effective luminaires [11], making OWP attractive for navigation and asset tracking in malls, airports, and museums where GNSS is unavailable, as well as in electromagnetic-sensitive environments such as hospitals and airplane cabins [10]. Moreover, the same LED infrastructure can provide Optical Wireless Communication (OWC), often termed Li-Fi [12], so that illumination, data, and positioning coexist in a unified platform, albeit with practical constraints on illumination and LOS availability.

Position estimation in OWP follows two paradigms: data-driven and model-based. Data-driven methods include fingerprinting with k-nearest neighbors; support vector machines/regression; random forests and gradient boosting; Gaussian process regression; multilayer perceptrons; convolutional neural networks for camera or structured PD signals; recurrent/temporal models (LSTM/GRU/Transformers); and autoencoder- or metric-learning-based embeddings for domain adaptation [13], [14]. While effective with abundant labeled data, these approaches are often opaque, sensitive to domain shift (hardware, illumination, occlusion), and require periodic retraining, offering limited physical guarantees [15]. By contrast, model-based estimators grounded in radiometry and geometry (e.g., Lambertian propagation, field-of-view

TABLE I
PERFORMANCE COMPARISON OF SINGLE-LED VLP SYSTEMS.

Year	Work	Configuration	Rx Orientation	2D/3D	Method	Error (APE)	Room (m^3)	Notes
2020	[17]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Model-based (Geometrical Equations)	2.52 cm	$1 \times 1 \times 1.5$	Sim.
2021	[18]	1 LED / 3 tilted PD	Fixed	2D	Long Short Term Memory- Fully Connected Network	0.92 cm	$1 \times 1.1 \times 1.75$	Exp.
2021	[19]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	2D	Bayesian algorithm	9.27 cm	$3 \times 3 \times 3$	Exp.
2022	[20]	1 LED / 1 Rotatable PD	Fixed	2D	Extreme Learning Machine	1.74 cm	$4 \times 4 \times 3.1$	Sim.
2023	[21]	1 LED / 4 PD	Fixed	2D	Least Square	7 cm (CDF _{90%})	$6 \times 6 \times 3$	Exp.
2024	[22]	1 reorientable LED / 1 PD	Arbitrary	2D	Model-based (Power Ratio)	5.9 cm (CDF _{90%})	$4 \times 4 \times 2.5$	Exp.
2024	[23]	1 LED / 4 PD	Fixed	2D	Extreme Learning Machine / WKNN	2.93 cm	$5 \times 5 \times 3$	Sim.
2024	[24]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Deep Residual Shrinkage method	5.85 cm	$3.6 \times 3.6 \times 3.0$	Exp.
2024	[25]	1 LED / 1 PD (Single-Tilted-Rotatable PD)	Rotatable	2D	Least Squares + Improved Grey Wolf Optimizer (LS-IGWO)	1.65 cm	$0.5 \times 0.5 \times 0.3$	Exp.
2024	[26]	1 LED / 4 PD + IMU (orientation)	Arbitrary	3D	MLP	1.77 cm	$1.5 \times 1.5 \times 1.5$	Exp.
2025	[27]	1 LED / 1 Rotatable PD	Fixed	2D	Least Square	4.96 cm	$5 \times 5 \times 4$	Exp.
2025	[28]	1 LED / 16 IRS / 1 PD	Fixed	2D	Levenberg-Marquardt algorithm	5.6 cm (RMSE)	$4 \times 5 \times 5$	Sim.
2025	Ours	1 reorientable LED / 1 PD	Arbitrary	3D	GLS	XXX	$3 \times 3 \times 2$	Sim.

Note: “Sim.” denotes simulation results; “Exp.” denotes experimental results.

constraints, pose kinematics) provide interpretability, sample efficiency, analyzable uncertainty, and reliable extrapolation, and they support joint system co-design [16].

Most existing OWP implementations rely on traditional multi-LED transmitter constellations, where multiple fixed LEDs with known positions act as anchors [29]. Using multiple LEDs allows the receiver to compute its position via triangulation, trilateration, or multilateration, analogous to how GNSS uses many satellites [10]. However, this multi-LED requirement complicates practical deployment. It presumes a dense infrastructure; at least four LEDs must be concurrently visible to the receiver to resolve a 3D position [30]–[33], which may not be true in environments with only sparse lighting. Installing and maintaining a dense LED network, with unique IDs or modulation for each lamp, increases installation cost and complexity [34]. Consequently, reducing the infrastructure burden by localizing with single light sources is a highly desirable goal in the field [13].

Recent research has explored OWP schemes that break from the multi-LED paradigm. One notable direction is single-LED positioning, which aims to achieve localization with only one transmitter [17]–[28]. A single light source dramatically simplifies infrastructure and calibration [35]. However, a lone transmitter cannot provide full positional information at a fixed orientation, since fundamentally one measurement is insufficient to resolve a 3D position. To overcome this, single-LED OWP systems incorporate additional cues or mechanisms in the UE side. Some approaches use additional sensors or diversity at the receiver to glean more information from the single source. While these single-LED solutions eliminate the need for multiple transmitters, they introduce their own complexities in the UE side, requiring user participation (device rotation), specialized receiver hardware (camera or multi-PD array), or additional aiding sensors like inertial measurement units. Moreover, some early single-LED demonstrations focused only on 2D positioning, assuming the receiver was on a fixed plane or height, which limits applicability. These limitations motivate the search for a new 3D single-LED positioning framework.

In parallel, recent advances in OWC have delivered agile beam-steering transmitters, ranging from high-speed MEMS micromirror gimbals [36], through liquid-crystal or metasurface deflectors that widen the field-of-view without sacrificing speed [37], to compact silicon-photonics optical-phased arrays that shift their beam tens of degrees in micro-seconds with no moving parts [38]. These demonstrations confirm that equipping a single indoor luminaire with a steering mechanism is feasible and highly synergistic.

In this work, we study a novel beam-steered single-LED single-PD OWP architecture for 3D indoor localization. The approach follows a two-stage pipeline—(i) direction finding from K steered orientations, and (ii) range recovery using a high-SNR beam-aligned measurement—without requiring receiver rotation or PD arrays. The main contributions are summarized as follows:

- A novel single-LED single-PD 3D OWP framework that achieves full 3D localization without receiver rotation, camera sensing, or PD arrays, enabled by beam steering.
- CRLB/PEB analysis for a composite observation model. We derive Fisher-information expressions and the associated position-error bound (PEB) for the joint set of K orientation-averaged received powers plus one beam-aligned range statistic, and we characterize identifiability and FIM rank conditions as a function of K .
- GA-based orientation-set design. We cast the transmitter orientation selection as a genetic-algorithm optimization that minimizes the 90th-percentile PEB over a 3D testbed, yielding practical design rules for the choice of K and set of orientations.
- Closed-form linear direction estimators. Using a ratio-based linearization, we develop a statistically efficient GLS estimator that accounts for heteroscedastic and correlated ratio errors, alongside a lightweight WLS approximation suitable for low-power implementations.

This paper is organized as follows. Section II presents the system model and the two-stage 3D localization procedure. Section III derives the position-error bound (PEB) for single-

LED positioning. Section IV optimizes the transmitter beam-orientation set. Section V develops a nonlinear (NL) direction estimator. Section VI introduces closed-form linear estimators (GLS and WLS). Section VII reports simulation results. Section VIII concludes the paper.

Notation: Bold lowercase letters denote column vectors (e.g., $\mathbf{x}$) and bold uppercase letters denote matrices (e.g., $\mathbf{A}$); scalars are in lightface. The Euclidean norm is $\|\cdot\|$, the inner product is $\mathbf{a} \cdot \mathbf{b}$, the trace is $\text{tr}(\cdot)$, and the gradient with respect to $\mathbf{r}$ is $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^T$. The transpose is $(\cdot)^T$, the identity and zero matrices are $\mathbf{I}_n$ and $\mathbf{0}$, respectively, and $\text{diag}(\cdot)$ forms a diagonal matrix from its arguments. The smallest eigenvalue and a corresponding unit-norm eigenvector of a symmetric matrix $\mathbf{M}$ are denoted by $\lambda_{\min}(\mathbf{M})$ and $\mathbf{v}_{\min}(\mathbf{M})$; the Loewner order $\succeq$ indicates positive semidefiniteness. Calligraphic symbols (e.g., $\mathcal{R}$) denote sets and $|\mathcal{A}|$ their cardinality; $\mathbb{R}^n$ is the n -dimensional real space and $\mathbb{S}^2$ the unit sphere in $\mathbb{R}^3$. Random variables follow the convention $x \sim \mathcal{N}(\mu, \sigma^2)$, with expectation $\mathbb{E}[\cdot]$ and variance $\text{Var}[\cdot]$. A hat $\hat{(\cdot)}$ denotes an estimate and an overbar $\bar{(\cdot)}$ a sample mean. The natural logarithm is $\ln(\cdot)$ and the base-10 logarithm is $\log_{10}(\cdot)$; signal-to-noise ratios in decibels use $10 \log_{10}(\cdot)$.

II. SYSTEM MODEL AND PROPOSED 3D LOCALIZATION METHOD

A. System Model

Fig. 1 illustrates the single-LED single-PD OWP setup analyzed in this work. A LED transmitter is mounted on the ceiling at a known position $\mathbf{t} = [0, 0, H]^T$ and can steer its optical axis along a unit vector $\mathbf{n}_t^{(i)} \in \mathbb{S}^2$ for the i -th orientation, $i = 1, \dots, K$. The receiver is a single PD with fixed normal along the vertical axis $\mathbf{n}_r = [0, 0, 1]^T$ located at an unknown position $\mathbf{r} = [x, y, z]^T$. Let $\mathbf{d} = \mathbf{r} - \mathbf{t}$ denote the transmitter–receiver displacement vector, $d = \|\mathbf{d}\|_2$ the separation distance, and $\mathbf{n}_d = \mathbf{d}/d$ the corresponding unit direction vector.

For the i -th LED orientation, the received signal strength (RSS) at the PD is modeled as:

$$S_r^{(i)} = R_{\text{pd}} P_t h_{\text{LOS}}^{(i)} + w_i, \quad (1)$$

where R_{pd} is the PD responsivity and $w_i \sim \mathcal{N}(0, \sigma_w^2)$ models a zero-mean additive white Gaussian noise (AWGN) term arising primarily from shot and Johnson noise, and assumed independent and identically distributed (i.i.d.) across orientations. Dividing (1) by R_{pd} yields the received optical power:

$$P_r^{(i)} = P_t h_{\text{LOS}}^{(i)} + n_i, \quad (2)$$

where P_t is the transmitted radiant power, $n_i \sim \mathcal{N}(0, \sigma^2)$ is zero-mean AWGN, remains i.i.d., with variance $\sigma^2 = (\sigma_w^2 / R_{\text{pd}}^2)$, and $h_{\text{LOS}}^{(i)}$ denotes the line-of-sight (LOS) channel gain, which for a Lambertian source is given by [39]:

$$h_{\text{LOS}}^{(i)} = \begin{cases} \frac{(m+1)A_{\text{det}}}{2\pi d^2} \cos^m(\phi_i) \cos(\psi), & \psi \leq \Psi_{\text{FOV}}, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where m is the Lambertian order of the LED, related to the semi-angle at half power $\Phi_{1/2}$ by $m = -\ln 2 / \ln(\cos(\Phi_{1/2}))$,

A_{det} the photodiode effective area, Ψ_{FOV} the receiver field of view, and the irradiance/incidence cosines geometrically given by:

$$\cos \phi_i = \frac{\mathbf{n}_t^{(i)} \cdot \mathbf{d}}{d}, \quad \cos \psi = \frac{\mathbf{n}_r \cdot (-\mathbf{d})}{d}. \quad (4)$$

At every orientation i the receiver acquires and averages N samples, producing the unbiased estimator of the received power:

$$\bar{P}_r^{(i)} = \frac{1}{N} \sum_{k=1}^N P_{r,k}^{(i)} \quad (5)$$

The corresponding instantaneous signal-to-noise ratio (SNR) and its testbed average are:

$$\text{SNR}_r^{(i)} = \frac{(R_{\text{pd}} P_t h_{\text{LOS}}^{(i)})^2}{\sigma_w^2}, \quad (6a)$$

$$\overline{\text{SNR}}_r = \frac{1}{K} \sum_{i=1}^K \text{SNR}_r^{(i)}, \quad (6b)$$

$$\overline{\text{SNR}} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \overline{\text{SNR}}_r. \quad (6c)$$

where $\mathcal{R}$ denotes the set of receiver testbed positions. Results are reported in decibels through $10 \log_{10}(\overline{\text{SNR}})$.

B. General 3D single-LED Positioning Procedure

The proposed localisation pipeline comprises two consecutive stages.

1) *Direction finding:* The first stage estimates the direction vector $\mathbf{n}_d$ by steering the LED through the K predefined orientations $\{\mathbf{n}_t^{(i)}\}_{i=1}^K$ and processing the power means $\{\bar{P}_r^{(i)}\}_{i=1}^K$. Section V (NL estimator) and Section VI (linear GLS/WLS estimators) detail statistically grounded solutions.

2) *Distance recovery:* After direction finding, the LED is aligned with the estimated direction $\hat{\mathbf{n}}_d$ and, for beam-forming symmetry, the receiver is re-oriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$. Under this beam-steered configuration and using (2), (3), and (4), the received power obeys:

$$P_r^* = \frac{C}{d^2} + n, \quad (7)$$

where C is an optical constant calibrated once per hardware set-up:

$$C = \frac{P_t(m+1)A_{\text{det}}}{2\pi}, \quad (8)$$

Collecting N independent measurements $\{P_{r,j}^*\}_{j=1}^N$ and using (5) yields the Minimum Variance Unbiased (MVU) estimate $\bar{P}_r^* \sim \mathcal{N}(\mu^*, \sigma^2/N)$. Substituting $\bar{P}_r^*$ into (7) provides the distance estimate:

$$\hat{d} = \sqrt{\frac{C}{\bar{P}_r^*}}, \quad (9)$$

and, finally, the 3D position estimate:

$$\hat{\mathbf{r}} = \mathbf{t} + \hat{d} \hat{\mathbf{n}}_d. \quad (10)$$

III. POSITION ERROR BOUND

This section derives the position error bound (PEB), i.e., the Cramér–Rao lower bound (CRLB) specialised to the receiver’s 3D coordinate vector $\mathbf{r} = [x, y, z]^T$. The PEB quantifies the minimum achievable variance of any unbiased estimator of $\mathbf{r}$ under the proposed OWP model and serves both as a benchmark for estimator performance and as an objective for system design.

A. Joint statistical model of the observations

According to Section II, the sample-mean optical power, as defined in (5), for each of the K transmitter orientations used during the direction finding stage is distributed as:

$$\bar{P}_r^{(i)} \sim \mathcal{N}(\mu_i(\mathbf{r}), \sigma^2/N), \quad i = 1, \dots, K, \quad (11)$$

where $\mu_i(\mathbf{r}) = P_t h_{\text{LOS}}^{(i)}(\mathbf{r})$ and $h_{\text{LOS}}^{(i)}$ is defined in (3) with the cosines in (4). During the distance recovery stage, the mean optical power in aligned configuration obeys:

$$\bar{P}_r \sim \mathcal{N}(\mu_{K+1}(\mathbf{r}), \sigma^2/N), \quad (12)$$

where $\mu_{K+1}(\mathbf{r}) = C/d^2(\mathbf{r})$, $d(\mathbf{r}) = \|\mathbf{r} - \mathbf{t}\|$ and C is the optical constant introduced in (8).

Stacking these $K + 1$ independent observations into $\mathbf{z} = [\bar{P}_r^{(1)}, \dots, \bar{P}_r^{(K)}, \bar{P}_r]^T$ yields the multivariate Gaussian model:

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{r}), (\sigma^2/N) \mathbf{I}_{K+1}), \quad (13)$$

with mean vector $\boldsymbol{\mu}(\mathbf{r}) = [\mu_1(\mathbf{r}), \dots, \mu_{K+1}(\mathbf{r})]^T$ and diagonal covariance σ^2/N .

B. Log-likelihood and score function

Under the assumption of independent Gaussian measurements, the log-likelihood of $\mathbf{r}$ given $\mathbf{z}$ is:

$$\ell(\mathbf{r}) = \ln L(\mathbf{r} | \mathbf{z}) = -\frac{N}{2\sigma^2} \sum_{i=1}^{K+1} [z_i - \mu_i(\mathbf{r})]^2 + c_0, \quad (14)$$

where c_0 is an additive constant independent of $\mathbf{r}$. Differentiating $\ell(\mathbf{r})$ with respect to $\mathbf{r}$ defines the score vector:

$$\mathbf{s}(\mathbf{r}) = \nabla_{\mathbf{r}} \ell(\mathbf{r}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} [z_i - \mu_i(\mathbf{r})] \nabla_{\mathbf{r}} \mu_i(\mathbf{r}), \quad (15)$$

where $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^T$ is the gradient operator.

C. Fisher Information Matrix

For distributions in the exponential family, such as the homoscedastic Gaussian model adopted here, the Fisher Information Matrix (FIM) can be obtained as the variance of the score vector, avoiding second-order differentiation and yielding greater numerical stability [40]:

$$\mathbf{I}(\mathbf{r}) = \text{Var}[\mathbf{s}(\mathbf{r})] = \mathbb{E}_{\mathbf{z}}[\mathbf{s}(\mathbf{r}) \mathbf{s}(\mathbf{r})^T] \quad (16)$$

$$= \frac{N}{\sigma^2} \sum_{i=1}^{K+1} [\nabla_{\mathbf{r}} \mu_i(\mathbf{r})] [\nabla_{\mathbf{r}} \mu_i(\mathbf{r})]^T. \quad (17)$$

The second equality follows from $\mathbb{E}[z_i] = \mu_i(\mathbf{r})$, which implies $\mathbb{E}[(z_i - \mu_i)(z_j - \mu_j)] = 0$ for $i \neq j$. Hence the total information is the coherent superposition of the differential contributions from each LED orientation and the range measurement. Required gradients are:

1) *Direction-finding measurements*: For the i -th orientation:

$$\nabla_{\mathbf{r}} \mu_i(\mathbf{r}) = P_t \nabla_{\mathbf{r}} h_{\text{LOS}}^{(i)}(\mathbf{r}), \quad (18)$$

which, by differentiating (3) via the chain rule on $d(\mathbf{r})$, $\cos \phi_i$ and $\cos \psi$, yields the closed-form

$$\nabla_{\mathbf{r}} \mu_i(\mathbf{r}) = \frac{C}{d^3} \left[m \cos^{m-1} \phi_i \cos \psi \mathbf{n}_t^{(i)} - \cos^m \phi_i \mathbf{n}_r - (m+3) \cos^m \phi_i \cos \psi \mathbf{n}_d \right], \quad (19)$$

which vanishes whenever $\psi > \Psi_{\text{FOV}}$ since $h_{\text{LOS}}^{(i)} = 0$ outside the receiver’s field of view.

2) *Distance-recovery measurement*:

$$\nabla_{\mathbf{r}} \mu_{K+1}(\mathbf{r}) = -\frac{2C}{d^4} (\mathbf{r} - \mathbf{t}) = -\frac{2C}{d^3} \mathbf{n}_d, \quad (20)$$

where $\mathbf{n}_d$ is the unit direction vector defined in Section II.

D. Formal definition of the PEB

Let $\hat{\mathbf{r}}$ be any unbiased estimator of $\mathbf{r}$. The Cramér–Rao inequality asserts:

$$\text{Cov}[\hat{\mathbf{r}}] \succeq \mathbf{I}^{-1}(\mathbf{r}). \quad (21)$$

Accordingly, a scalar performance metric, the PEB, is defined as:

$$\text{PEB}(\mathbf{r}) = \sqrt{\text{tr}\{\mathbf{I}^{-1}(\mathbf{r})\}}. \quad (22)$$

The square root of the trace of $\mathbf{I}^{-1}$ corresponds to the minimum achievable root-mean-square error (RMSE) in position estimation for the described OWP configuration. A small PEB indicates that the chosen set of LED orientations and the range measurement provide sufficient and complementary information for high-precision localisation; conversely, a large PEB suggests the need to increase K , improve the SNR, or redesign the orientation pattern.

E. Number of LED Orientations

1) *Underdetermined regime* ($K \leq 2$): In this regime, the Fisher information matrix $\mathbf{I}(\mathbf{r})$ from (17) is rank-deficient. Each of the K orientation measurements contributes at most to one independent gradient vector, and the distance-recovery measurement adds a single additional vector. When $K \leq 2$, there are fewer than three linearly independent gradients in total, so $\mathbf{I}(\mathbf{r})$ is singular and the CRLB on any unbiased position estimate diverges [29].

2) *Just-determined regime* ($K = 3$): With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the 3D parameter space. In generic geometries the FIM is full rank and $\mathbf{I}^{-1}(\mathbf{r})$ exists, yielding a finite PEB. However, the absence of redundancy makes the bound highly sensitive to small perturbations in orientation geometry [18].

3) *Overdetermined regime* ($K \geq 4$): For four or more orientations, the FIM accumulates redundant outer-products of gradients. This overdetermination improves the numerical conditioning of $\mathbf{I}(\mathbf{r})$ and generally lowers the PEB, since extra measurements provide additional independent information beyond the minimal requirement [10], [26].

IV. OPTIMIZATION OF THE ORIENTATION SET

Equation (19) shows that the FIM $\mathbf{I}(\mathbf{r})$ depends explicitly on the transmitter orientation vectors $\{\mathbf{n}_t^{(i)}\}_{i=1}^K$, and hence so does the PEB in (22). Accordingly, selecting the orientation set that minimises the PEB is essential for optimal localisation accuracy. Since $\text{PEB}(\{\mathbf{n}_t^{(i)}\})$ is non-convex in the orientation parameters, direct analytical minimisation is intractable. To address this, we employ a genetic algorithm (GA) to search for the orientation ensemble that minimises the global PEB for a given K .

TABLE II
PARAMETERS USED FOR GA OPTIMIZATION.

System Parameters	
Parameter	Value
Transmitter position	$T = (0, 0, 2)$ m
Transmitted optical power	$P_t = 0.405$ W
Half-power semi-angle	$\Phi_{1/2} \in \{30^\circ, 45^\circ, 60^\circ\}$
Lambertian order	$m = -\ln 2 / \ln(\cos(\Phi_{1/2}))$
Photodiode effective area	$A_{\text{det}} = 4.8 \times 5.5$ mm ²
Receiver field of view (FOV)	$\Psi_{\text{FOV}} = 85^\circ$
Electrical noise variance per sample	$\sigma_w^2 = 1.19 \times 10^{-14}$ A ²
PD responsivity	$R_{\text{pd}} = 0.63$ A/W
Noise variance per sample	$\sigma^2 = 3 \times 10^{-14}$ W ²
Samples per orientation	$N = 1000$
Genetic Algorithm	
Parameter	Value
Initial population	300
Maximum generations	200
Crossover rate	0.8
Search range	$\theta_i \in [0, 80^\circ]$, $\rho_i \in [0, 360^\circ]$
Optimization metric	RMSE
Decision variables	$[\theta_1, \rho_1, \dots, \theta_K, \rho_K]$
3D testbed	
Parameter	Value
Room dimensions (LxWxH)	$3 \times 3 \times 2$ m ³
Range in x, y	$[-1.5, 1.5]$ m
Range in z	$[0, 1.2]$ m
Grid step	0.2 m
Total number of points tested	1792

Each GA individual encodes K steering directions via tilt–azimuth pairs $\{(\theta_i, \rho_i)\}_{i=1}^K$, yielding a $2K$ -dimensional genotype $\mathbf{x} = [\theta_1, \rho_1, \dots, \theta_K, \rho_K]^T$. We adopt a nadir-referenced spherical parameterisation where θ_i is the tilt from $-\mathbf{u}_z$ and ρ_i is the azimuth in the x – y plane measured counterclockwise from $+\mathbf{u}_x$, with domains $\theta_i \in [0, \theta_{\text{max}}]$ and $\rho_i \in [-0^\circ, 360^\circ]$. The mapping to unit-norm orientation vectors in the reference frame $(\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z)$ is:

$$\mathbf{n}_t^{(i)} = (\sin \theta_i \cos \rho_i, \sin \theta_i \sin \rho_i, -\cos \theta_i). \quad (23)$$

Spherical coordinates are preferred to Cartesian because they describe unit-norm directions with two angles, avoiding the nonlinear unit-norm constraint and providing bounded domains that simplify GA search.

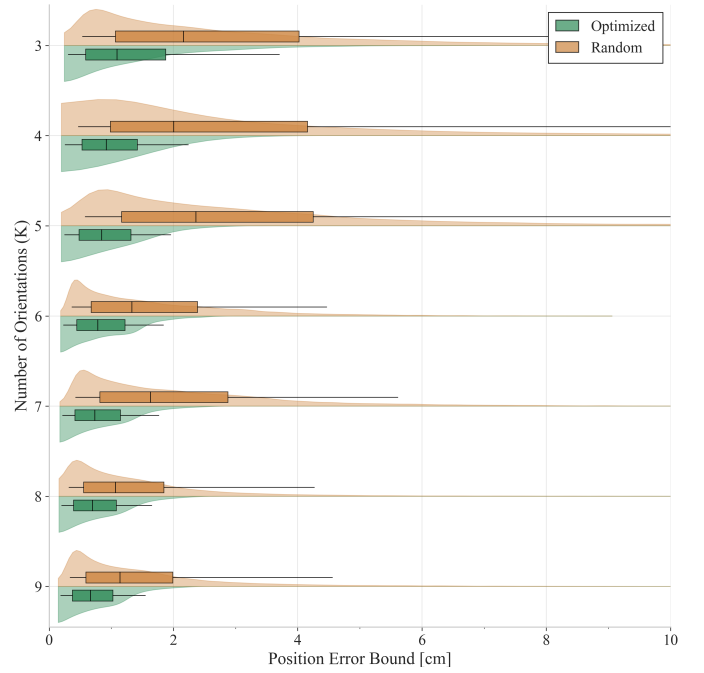


Fig. 2. RMSE performance in a 3×3 m testbed using an optimal set of K orientations versus a sub-optimal set of K orientations.

The GA's fitness metric is defined as the RMSE of the PEB over all 3D grid points spanning the testbed, computed by evaluating $\text{PEB}(\mathbf{r})$ from (22). Standard genetic operators such as tournament selection, crossover probability p_c , and mutation rate p_m , are applied over G generations. The algorithm is implemented in MATLAB; optimization parameters are summarised in Table II.

Table III lists the optimal K -orientation sets for the system parameters in Table II. Each set is unique and no set is a subset of another. For every optimal orientation ensemble, we compute the PEB (or equivalently the RMSE error) at each receiver location within the testbed grid. In Fig. 2, the box-and-whisker plots for the optimal sets are shown in green, alongside those for randomly chosen orientations. Quantitatively, the optimized sets achieve an average PEB below 1.5 cm for all $K \in \{3, \dots, 9\}$ (worst mean 1.43 cm at $K = 3$) and reach 0.74 cm at $K = 9$ (a 49%–72% reduction relative to random). These gains are reflected in the distributions, where the optimized sets exhibit lower medians and interquartile ranges, with the central tendency improving monotonically as K increases.

Figure 3 presents spatial heat maps of the PEB across a 3×3 m floor at $z = 0.8$ m for $K = 5$ orientations with the transmitter at the room centre. In the optimal configuration (Fig. 3a), the highest errors occur uniformly along the boundaries and especially at the corners, analogous to multi-transmitter layouts, but remain spatially consistent. By contrast, the random configuration (Fig. 3b) exhibits both larger and more irregular error patterns.

Figure 4 depicts the 90th-percentile of the PEB distribution ($\text{PEB}_{90\%}$) for the optimal K -orientation set under three $\Phi_{1/2}$ scenarios at an effective SNR of 14 dB. In all scenarios,

TABLE III
OPTIMAL ORIENTATIONS (θ_i, ρ_i) IN DEGREES $[\circ]$ FOR VARIOUS K IN A $3 \times 3 \times 2 \text{ m}^3$ ROOM

K	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$	$i = 8$	$i = 9$
3	(35.4, 140.1)	(33.3, 36.4)	(29.6, 262.7)	-	-	-	-	-	-
4	(38.9, 90.6)	(41.5, 0.10)	(41.8, 180.1)	(38.8, 270.2)	-	-	-	-	-
5	(0.1, 211.1)	(50.7, 180.0)	(50.4, 359.9)	(50.6, 90.0)	(50.6, 270.0)	-	-	-	-
6	(17.2, 306.9)	(54.5, 266.1)	(22.5, 140.4)	(52.2, 360.0)	(52.4, 84.0)	(55.8, 185.2)	-	-	-
7	(5.4, 305.9)	(53.8, 170.7)	(27.8, 43.8)	(58.9, 355.7)	(54.4, 96.5)	(35.1, 220.0)	(54.8, 278.6)	-	-
8	(6.5, 318.3)	(61.5, 268.3)	(27.3, 317.0)	(51.8, 89.4)	(57.8, 5.80)	(53.7, 171.3)	(38.0, 200.3)	(39.3, 91.1)	-
9	(2.50, 28.2)	(36.5, 266.8)	(33.9, 182.3)	(56.9, 178.7)	(42.2, 78.4)	(53.1, 97.5)	(57.9, 359.7)	(37.1, 355.1)	(58.2, 272.1)

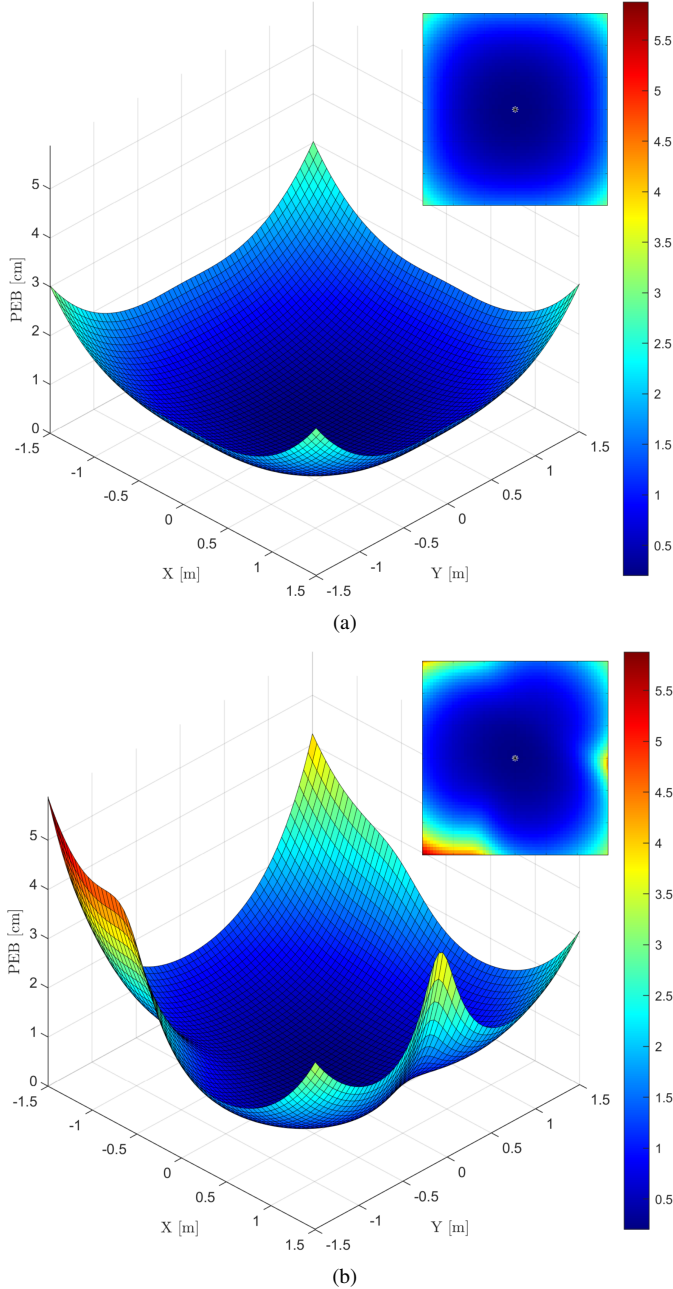


Fig. 3. Heat map of the PEB across a $3 \times 3 \text{ m}$ testbed at $z = 0.8 \text{ m}$ for $K = 5$ orientations: (a) optimal set, (b) random set.

PEB_{90%} decreases as K increases, indicating improved localisation precision with additional orientations. A balance among acquisition latency, computational load, and positioning accuracy is attained at $K = 5$. For $\Phi_{1/2} = 30^\circ$, at least four orientations are necessary due to the increased directivity limiting coverage with only three orientations; configurations with $K \geq 4$ outperform alternatives. In contrast, $\Phi_{1/2} = 45^\circ$ achieves competitive error performance starting at $K = 3$, and given its widespread use in commercial LEDs, it is adopted for the subsequent simulations.

Figure 5 illustrates the dependence of the PEB_{90%} on the SNR for various optimized K -orientation sets. As the SNR increases, PEB_{90%} decreases monotonically. Moreover, at any fixed SNR, increasing K yields a lower PEB, and the spacing between the curves diminishes at high SNR, indicating reduced incremental benefit from additional orientations under noise-limited conditions.

V. NONLINEAR ESTIMATOR

The optical power collected by the PD depends nonlinearly on its relative position to the LED, through the range d , the irradiance angle ϕ_i , and the incidence angle ψ . Denoting the receiver's orientation vector as $\mathbf{n}_r = [\alpha, \beta, \gamma]^T$, we have

$$\cos \psi = \mathbf{n}_r \cdot (-\mathbf{n}_d) = -\frac{\alpha x + \beta y + \gamma(z - H)}{d}, \quad (24)$$

where $\mathbf{r} = (x, y, z)^T$ is the receiver position and H the LED height. Substituting (4), (24), and $d = \sqrt{x^2 + y^2 + (z - H)^2}$ into the received-power model (2), and then forming the sample mean as in (5), yields:

$$\bar{P}_r^{(i)} = -C \frac{(a_i x + b_i y + c_i(z - H))^m}{[x^2 + y^2 + (z - H)^2]^{\frac{m+3}{2}}} (\alpha x + \beta y + \gamma(z - H)) + n, \quad (25)$$

where $n \sim \mathcal{N}(0, \sigma^2/N)$ and C is defined in (8). Thus, the observations relate nonlinearly to the receiver coordinates $\mathbf{r}$. To recover the direction unit vector $\mathbf{n}_d$ rather than the full coordinate triple, one imposes the constraint $x^2 + y^2 + (z - H)^2 = 1$.

Defining:

$$Q_i(x, y, z) = a_i x + b_i y + c_i(z - H), \quad (26)$$

$$L(x, y, z) = \alpha x + \beta y + \gamma(z - H), \quad (27)$$

the model in (25) can be compactly written as:

$$\bar{P}_r^{(i)} = -C Q_i^m(x, y, z) L(x, y, z) + n. \quad (28)$$

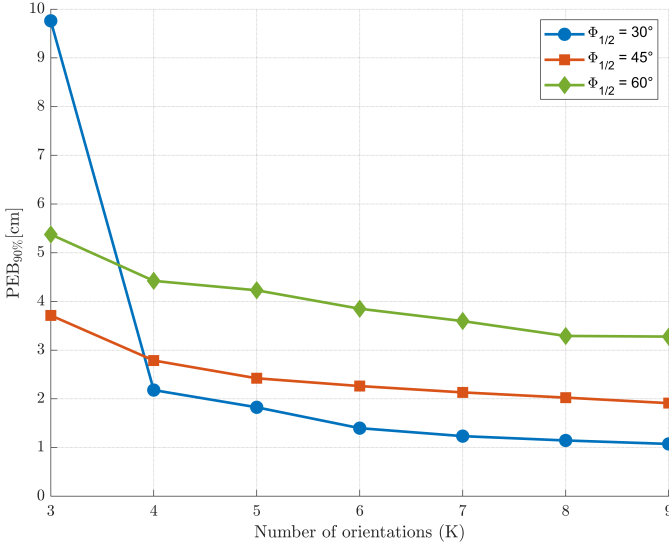


Fig. 4. PEB_{90%} versus number of orientations under three $\Phi_{1/2}$ scenarios with GA-optimised sets of orientations.

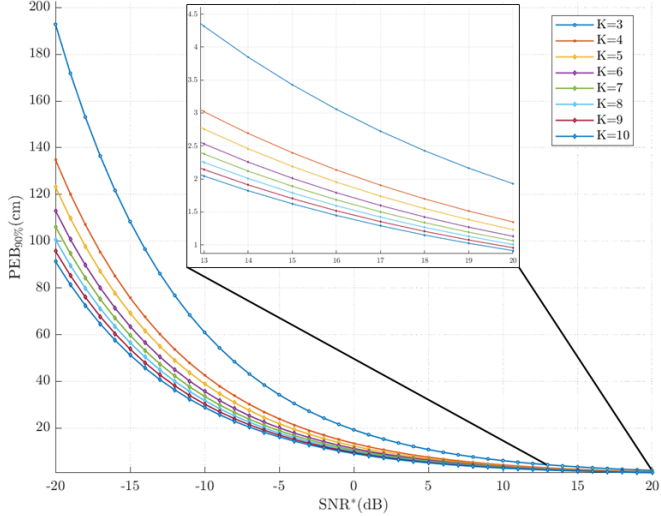


Fig. 5. Variation of the PEB_{90%} with $\overline{\text{SNR}}$ for different optimized K -orientation configurations.

This scenario exemplifies a classical estimation problem in which the averaged observations $\bar{P}_r^{(i)}$ for each orientation i are nonlinearly related to the unknown receiver coordinates and corrupted by additive Gaussian noise of known variance. Although no unbiased efficient estimator exists for this NL model, one can derive a MVU estimate by solving a constrained NL least-squares problem.

Define the cost function:

$$F(x, y, z) = \sum_{i=1}^K F_i(x, y, z), \quad (29)$$

where:

$$F_i(x, y, z) = [\bar{P}_r^{(i)} + C Q_i(x, y, z) L(x, y, z)]^2. \quad (30)$$

Here Q_i and L are as defined in (26)–(27), and C is the optical constant introduced in (7).

Since our goal is to recover the unit-norm direction vector $\mathbf{n}_d$ rather than the absolute coordinates, the variables $(x, y, z - H)$ must lie on the unit sphere $\mathbb{S}^2$. Moreover, the polynomial terms $Q_i(x, y, z)$ and $L(x, y, z)$ must satisfy:

$$Q_i(x, y, z) \geq 0, \quad L(x, y, z) \leq 0, \quad (31)$$

corresponding respectively to $\cos \phi_i \geq 0$ and $\cos \psi \geq 0$ for incidence and irradiance angles in $[0^\circ, 90^\circ]$.

Thus, the NL direction-finding estimator is:

$$\hat{\mathbf{n}}_{d, \text{NL}} = \arg \min_{\substack{(x, y, z - H) \in \mathbb{S}^2 \\ Q_i(x, y, z) \geq 0 \\ L(x, y, z) \leq 0}} F(x, y, z), \quad (32)$$

with $F(x, y, z)$ given by (29)–(30). In practice, this optimisation can be performed in MATLAB using the `solve` function, which internally invokes `fmincon` and employs gradient-descent algorithms suitable for constrained NL problems.

VI. LINEAR ESTIMATOR

The present section develops the direction-finding stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalised least squares (GLS), which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares (WLS), an inexpensive approximation that neglects the weak cross-correlation between equations. For implementation details, an end-to-end pseudocode of the full pipeline (GLS/WLS direction finding, distance recovery, and 3D position estimation) is provided in Algorithm 1.

A. Direction Estimation via GLS

This subsection consolidates the analytical procedure that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{n}_d$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

1) *Lambertian power ratios and homogeneous constraints:* Let $\mu_i = \mathbb{E}[P_r^{(i)}]$ be the noise-free power at orientation i . We define power ratio β as:

$$\beta_i = (\mu_i / \mu_1)^{1/m}, \quad i = 2, \dots, K, \quad (33)$$

and, from (2), (3), (4) and (8):

$$\mu_i = -C \frac{(\mathbf{n}_t^{(i)} \cdot \mathbf{d})^m (\mathbf{n}_r \cdot \mathbf{d})}{d^{m+3}}, \quad (34)$$

using (34) in (33), each β_i imposes the orthogonality condition:

$$\mathbf{a}_i \cdot \mathbf{d} = 0, \quad i = 2, \dots, K, \quad (35)$$

i.e., $\mathbf{a}_i \perp \mathbf{d}$, with:

$$\mathbf{a}_i = \mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}, \quad (36)$$

therefore supplies $K - 1$ homogeneous hyperplanes that intersect at the true $\mathbf{d}$.

2) *Stochastic perturbation of the ratios*: Recall from (5) that the sample mean is:

$$\bar{P}_r^{(i)} = \frac{1}{N} \sum_k P_{r,k}^{(i)} \sim \mathcal{N}(\mu_i, \sigma^2/N). \quad (37)$$

For notational convenience, define $\hat{\mu}_i \triangleq \bar{P}_r^{(i)}$, which is the MVU estimator of μ_i . Injecting $\hat{\mu}_i$ into (33) yields:

$$\hat{\beta}_i = g_i(\hat{\mu}_1, \hat{\mu}_i), \quad g_i(x_1, x_2) = \left(\frac{x_2}{x_1} \right)^{\frac{1}{m}}. \quad (38)$$

A first-order Taylor expansion of g_i centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\top$ gives:

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1} \right), \quad (39)$$

Hence every $\hat{\beta}_i$ shares the common perturbation n_1 , producing correlation across orientations.

Define the ratio error:

$$\tilde{n}_i = \hat{\beta}_i - \beta_i. \quad (40)$$

Using (39) and the independence of n_i and n_1 ,

$$\text{Var}[\tilde{n}_i] = \frac{\sigma^2}{Nm^2} \beta_i^2 (\mu_i^{-2} + \mu_1^{-2}), \quad (41)$$

$$\text{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{Nm^2} \beta_i \beta_j \mu_1^{-2}, \quad i \neq j. \quad (42)$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (41) demonstrates that the variance of $\tilde{n}_i$ depends explicitly on μ_i , even though each n_i has the same variance σ^2 . In other words, a single-LED system exhibits *heteroscedastic* ratio errors.

3) *Residuals and their covariance*: Define the empirical constraint normals as $\hat{\mathbf{a}}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i \mathbf{n}_t^{(1)}$ and the corresponding residuals as:

$$\xi_i = \hat{\mathbf{a}}_i \cdot \mathbf{d}, \quad i = 2, \dots, K. \quad (43)$$

Using (40) and (36), we have $\hat{\mathbf{a}}_i = \mathbf{a}_i - \tilde{n}_i \mathbf{n}_t^{(1)}$. Since $\mathbf{a}_i \cdot \mathbf{d} = 0$ by (35), it follows that:

$$\xi_i = -(\mathbf{n}_t^{(1)} \cdot \mathbf{d}) \tilde{n}_i = -\kappa \tilde{n}_i, \quad \kappa = \mathbf{n}_t^{(1)} \cdot \mathbf{d}. \quad (44)$$

Stacking $\boldsymbol{\xi} = [\xi_2, \dots, \xi_K]^\top$ and $\tilde{\mathbf{n}} = [\tilde{n}_2, \dots, \tilde{n}_K]^\top$ yields:

$$\begin{aligned} \boldsymbol{\xi} &= -\kappa \tilde{\mathbf{n}}, \\ \mathbb{E}[\boldsymbol{\xi}] &= \mathbf{0}, \\ \boldsymbol{\Sigma}_\xi &= \text{Cov}(\boldsymbol{\xi}) = \kappa^2 \boldsymbol{\Sigma}_\beta, \end{aligned} \quad (45)$$

where $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{(K-1) \times (K-1)}$ collects the variances and covariances of $\tilde{n}_i$ given in (41)–(42).

4) *GLS formulation and closed-form solution*: Stack the empirical constraint normals into $\hat{\mathbf{A}} = [\hat{\mathbf{a}}_2 \cdots \hat{\mathbf{a}}_K] \in \mathbb{R}^{3 \times (K-1)}$. From (43)–(45), the stacked residuals satisfy $\boldsymbol{\xi} = \hat{\mathbf{A}}^\top \mathbf{d}$ with $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\xi)$ and $\boldsymbol{\Sigma}_\xi = \kappa^2 \boldsymbol{\Sigma}_\beta$, $\kappa = \mathbf{n}_t^{(1)} \cdot \mathbf{d}$. Given an observation of $\boldsymbol{\xi} = \hat{\mathbf{A}}^\top \mathbf{d}$, the maximum-likelihood (equivalently, GLS) estimate of the direction is obtained by minimizing the Mahalanobis norm of $\boldsymbol{\xi}$:

$$\begin{aligned} \hat{\mathbf{d}} &= \arg \min_{\|\mathbf{d}\|=1} \left\| \hat{\mathbf{A}}^\top \mathbf{d} \right\|_{\boldsymbol{\Sigma}_\xi^{-1}}^2 \\ &= \arg \min_{\|\mathbf{d}\|=1} \mathbf{d}^\top (\hat{\mathbf{A}} \boldsymbol{\Sigma}_\xi^{-1} \hat{\mathbf{A}}^\top) \mathbf{d}. \end{aligned} \quad (46)$$

Algorithm 1 3D Position Estimation from n Orientations

Require: Power samples $P_{r,k}^{(i)}$ ($k = 1, \dots, N$, $i = 1, \dots, K$);

steering vectors $\mathbf{n}_t^{(i)}$; Lambertian order m ;
noise variance σ^2 ; MODE $\in \{\text{GLS}, \text{WLS}\}$;
radiometric constant C .

Ensure: $\hat{\mathbf{r}}$ (3D position of the receiver)

```

1: Direction finding:
2:   Compute  $\hat{\mu}_i = \frac{1}{N} \sum_{k=1}^N P_{r,k}^{(i)}$  for  $i = 1, \dots, K$ 
3:   Compute  $\hat{\beta}_i = (\hat{\mu}_i / \hat{\mu}_1)^{1/m}$  for  $i = 2, \dots, K$ 
4:   Form  $\mathbf{a}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i \mathbf{n}_t^{(1)}$ 
5:   Set  $\mathbf{A} = [\mathbf{a}_2, \dots, \mathbf{a}_n]$ 
6:   if MODE = GLS then
7:     Build  $\boldsymbol{\Sigma}_\beta$  via (41)–(42), substituting  $\beta_i \leftarrow \hat{\beta}_i$ 
8:      $\mathbf{M} \leftarrow \mathbf{A} \boldsymbol{\Sigma}_\beta^{-1} \mathbf{A}^\top$ 
9:   else ▷ WLS (diagonal approximation)
10:     $\tilde{\boldsymbol{\Sigma}}_\beta \leftarrow \text{diag}\{\hat{\beta}_i^2 (\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2})\}$ 
11:     $\mathbf{M} \leftarrow \mathbf{A} \tilde{\boldsymbol{\Sigma}}_\beta^{-1} \mathbf{A}^\top$ 
12:   end if
13:   Compute  $\mathbf{v}_{\min}$  = eigenvector of  $\mathbf{M}$  with smallest eigenvalue
14:    $\hat{\mathbf{n}}_d \leftarrow \mathbf{v}_{\min} / \|\mathbf{v}_{\min}\|$ 
15:   if  $\hat{\mathbf{n}}_d \cdot \mathbf{n}_t^{(1)} < 0$  then
16:      $\hat{\mathbf{n}}_d \leftarrow -\hat{\mathbf{n}}_d$  ▷ Ensure pointing to Rx
17:   end if

18: Distance Estimation (beamsteering):
19:   Set  $\mathbf{n}_t \leftarrow \hat{\mathbf{n}}_d$ ,  $\mathbf{n}_r \leftarrow -\hat{\mathbf{n}}_d$ 
20:   Collect  $N$  power samples  $P_{r,k}^{*(k)}$ 
21:   Compute  $\hat{\mu}^* = \frac{1}{N} \sum_{k=1}^N P_{r,k}^{*(k)}$ 
22:    $\hat{d} \leftarrow \sqrt{\frac{C}{\hat{\mu}^*}}$ 

23: Final 3D position:
24:    $\hat{\mathbf{r}} \leftarrow \mathbf{t} + \hat{d} \hat{\mathbf{n}}_d$ 
25: return  $\hat{\mathbf{r}}$ 

```

Since $\boldsymbol{\Sigma}_\xi = \kappa^2 \boldsymbol{\Sigma}_\beta$, the unknown scale κ^2 cancels in (46); hence one may use $\boldsymbol{\Sigma}_\beta^{-1}$ in place of $\boldsymbol{\Sigma}_\xi^{-1}$ without affecting the minimizer. The problem is a constrained Rayleigh quotient whose solution is the unit eigenvector associated with the smallest eigenvalue of:

$$\mathbf{M}_{\text{GLS}} = \hat{\mathbf{A}} \boldsymbol{\Sigma}_\beta^{-1} \hat{\mathbf{A}}^\top \in \mathbb{R}^{3 \times 3}. \quad (47)$$

Therefore,

$$\hat{\mathbf{n}}_{d,\text{GLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})\|}, \quad \hat{\mathbf{n}}_{d,\text{GLS}} \cdot \mathbf{n}_t^{(1)} > 0, \quad (48)$$

where $\mathbf{v}_{\min}(\cdot)$ denotes the eigenvector corresponding to the smallest eigenvalue; the sign convention ensures the direction points from the transmitter toward the receiver.

B. WLS as a practical simplification

The GLS solution in (46) requires inverting the $(K-1) \times (K-1)$ covariance $\boldsymbol{\Sigma}_\xi = \kappa^2 \boldsymbol{\Sigma}_\beta$, typically via a Cholesky factorization [41]. For real-time, low-power receivers, a lighter alternative is to neglect the off-diagonal terms in (42) and approximate:

$$\tilde{\boldsymbol{\Sigma}}_\beta = \text{diag}\{\beta_i^2 (\mu_i^{-2} + \mu_1^{-2})\}_{i=2}^K. \quad (49)$$

Since $\boldsymbol{\Sigma}_\xi = \kappa^2 \boldsymbol{\Sigma}_\beta$, the unknown scale κ^2 cancels, and we define the WLS weighting matrix as $\mathbf{W} = \tilde{\boldsymbol{\Sigma}}_\beta^{-1}$.

Using the empirical normals $\hat{\mathbf{A}}$ from the previous subsection, the WLS criterion is the Rayleigh quotient:

$$J_{\text{WLS}}(\mathbf{d}) = \mathbf{d}^T (\hat{\mathbf{A}} \mathbf{W} \hat{\mathbf{A}}^T) \mathbf{d}, \quad \|\mathbf{d}\| = 1, \quad (50)$$

and we introduce the associated WLS matrix:

$$\mathbf{M}_{\text{WLS}} = \hat{\mathbf{A}} \mathbf{W} \hat{\mathbf{A}}^T \in \mathbb{R}^{3 \times 3}. \quad (51)$$

The minimizer is the unit eigenvector corresponding to the smallest eigenvalue of $\mathbf{M}_{\text{WLS}}$:

$$\hat{\mathbf{n}}_{d,\text{WLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})\|}, \quad \hat{\mathbf{n}}_{d,\text{WLS}} \cdot \mathbf{n}_t^{(1)} > 0. \quad (52)$$

The neglected cross-covariances in (42) scale with μ_1^{-2} , whereas the retained diagonal terms scale with $\mu_i^{-2} + \mu_1^{-2}$. Thus, when the reference orientation $i = 1$ is chosen such that $\mu_1 \gg \mu_i$ for most i , the off-diagonal terms are negligible relative to the diagonal terms and WLS closely tracks GLS while avoiding a full covariance inversion. In more symmetric geometries where $\mu_i \approx \mu_1$ for many i , the cross-correlations are non-negligible and the full GLS should be used for optimal performance.

VII. RESULTS

Figure 6 presents the cumulative distribution function (CDF) of the position estimation error for each algorithm introduced above, using optimal orientation sets of $K = 5$ and $K = 9$. The GLS estimator achieves performance comparable to the NL estimator and approaches the theoretical PEB, whereas the WLS estimator incurs higher error but requires significantly lower computational complexity, thereby reducing latency.

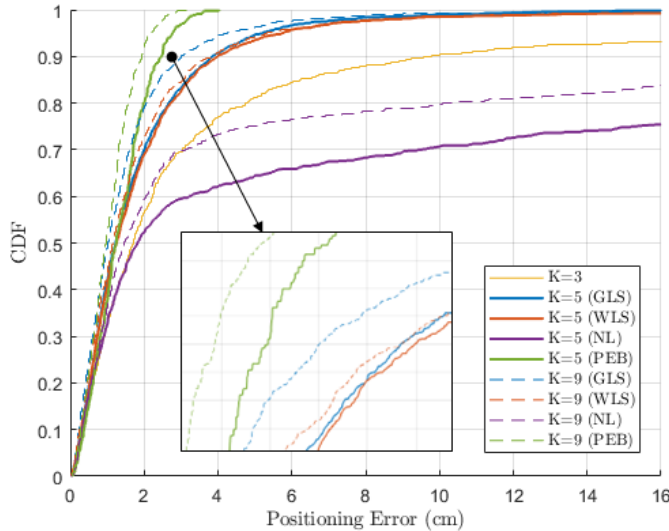


Fig. 6. Comparación de CDF de diferentes métodos y K .

Figure 7 compares the computation latency for $K = 5$ and $K = 9$. The NL estimator exhibits the greatest latency due to its iterative optimization procedure, while the closed-form GLS and WLS solutions execute more rapidly.

Figure 8 illustrates the 3D position estimates produced by WLS and GLS for $K = 5$ at three receiver heights $h = 0, 0.6, \text{ and } 1.2 \text{ m}$, plotted against the true reference positions.

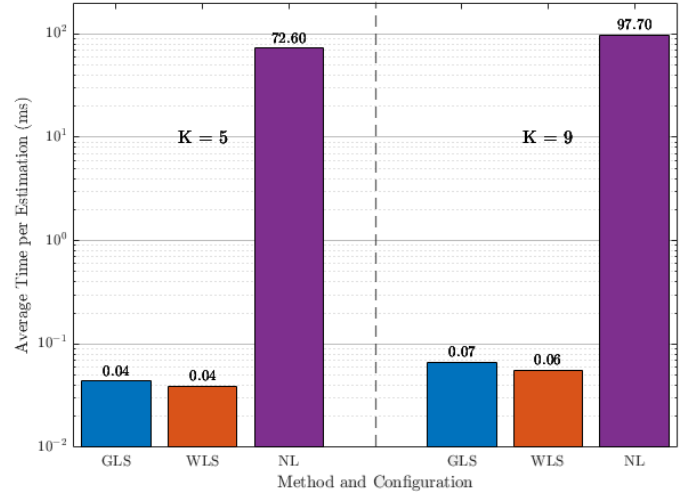


Fig. 7. Computation latency of the NL, GLS, and WLS estimators for $K = 5$ and $K = 9$.

At $h = 1.2 \text{ m}$, estimation errors increase near the testbed boundaries, particularly at the corners, consistent with the theoretical minimum PEB analysis (Fig. 3a) and attributable to reduced SNR at extreme incidence angles. In contrast, at $h = 0 \text{ m}$ the estimates exhibit greater spatial uniformity.

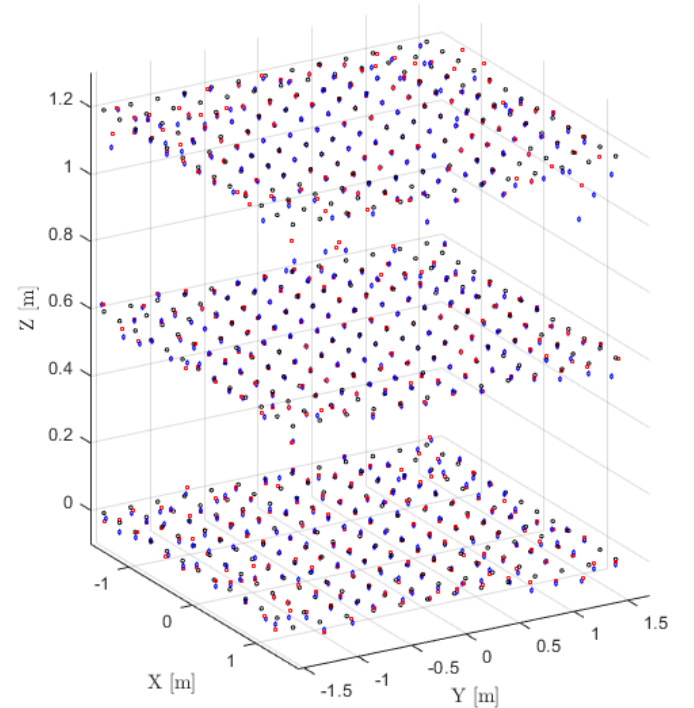


Fig. 8. 3D position estimates at three receiver heights for $K = 5$ orientations: GLS (blue), WLS (red), true positions (black).

VIII. CONCLUSION

This paper presented a novel beam-steered single-LED OWP system for accurate 3D positioning. The key contributions include: (1) a comprehensive mathematical framework incorporating both linear and NL estimators, (2) derivation

of the theoretical PEB for performance analysis, (3) genetic algorithm-based optimization of LED orientation sets, and (4) extensive performance evaluation.

The proposed system achieves centimeter-level accuracy while requiring minimal infrastructure compared to traditional multi-LED systems. The optimized 5-orientation configuration provides an excellent trade-off between positioning accuracy and system complexity. Future work will explore real-time implementation and experimental validation of the proposed algorithms.

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